Firewall Integration and Monitoring Report

Task 1: Firewall Integration and Monitoring (pfSense and Wazuh)
SOC Team, ITSOLERA



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This report documents the deployment and testing of a software firewall integrated with Wazuh for centralized log monitoring. The goal of this task was to set up pfSense as a firewall, connect it to a Wazuh SIEM for visibility, and create and test security rules that block traffic based on country of origin and by website domain. This document covers the firewall configuration, the rules that were created, the traffic that was generated to test those rules, the alerts and logs seen in Wazuh, and closing findings and recommendations.

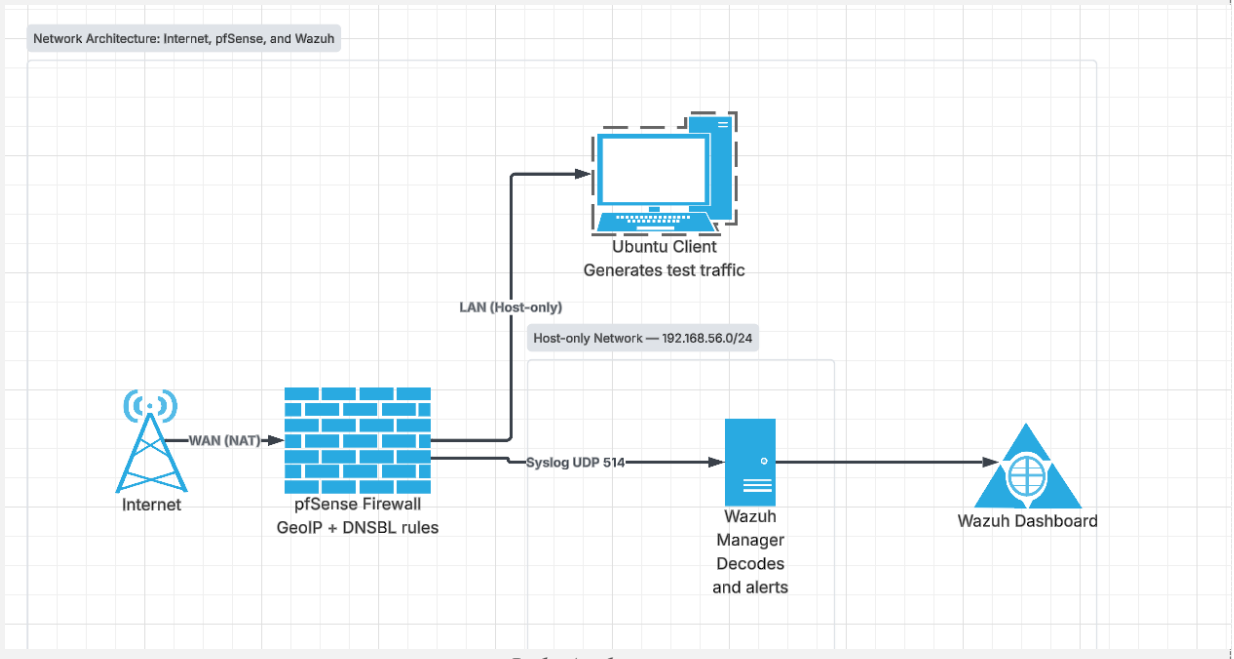
A. Firewall Configuration

1. Environment and Topology

The lab was built entirely using virtual machines in VirtualBox, running on a host machine. Three virtual machines were used for this task:

- pfSense (Community Edition) - acts as the firewall, sitting between the internet and the internal network.
- Wazuh Manager - receives logs from pfSense and generates alerts.
- Ubuntu client - used as a normal device on the internal network to generate test traffic.

pfSense was given two virtual network adapters. The WAN adapter was set to NAT, which lets pfSense reach the internet without being exposed on the real network. The LAN adapter was set to Host-only, which creates a private network shared only between the VMs on this host. This mirrors a real firewall setup where one side faces the outside world and the other side faces the protected internal network.



Lab Architecture

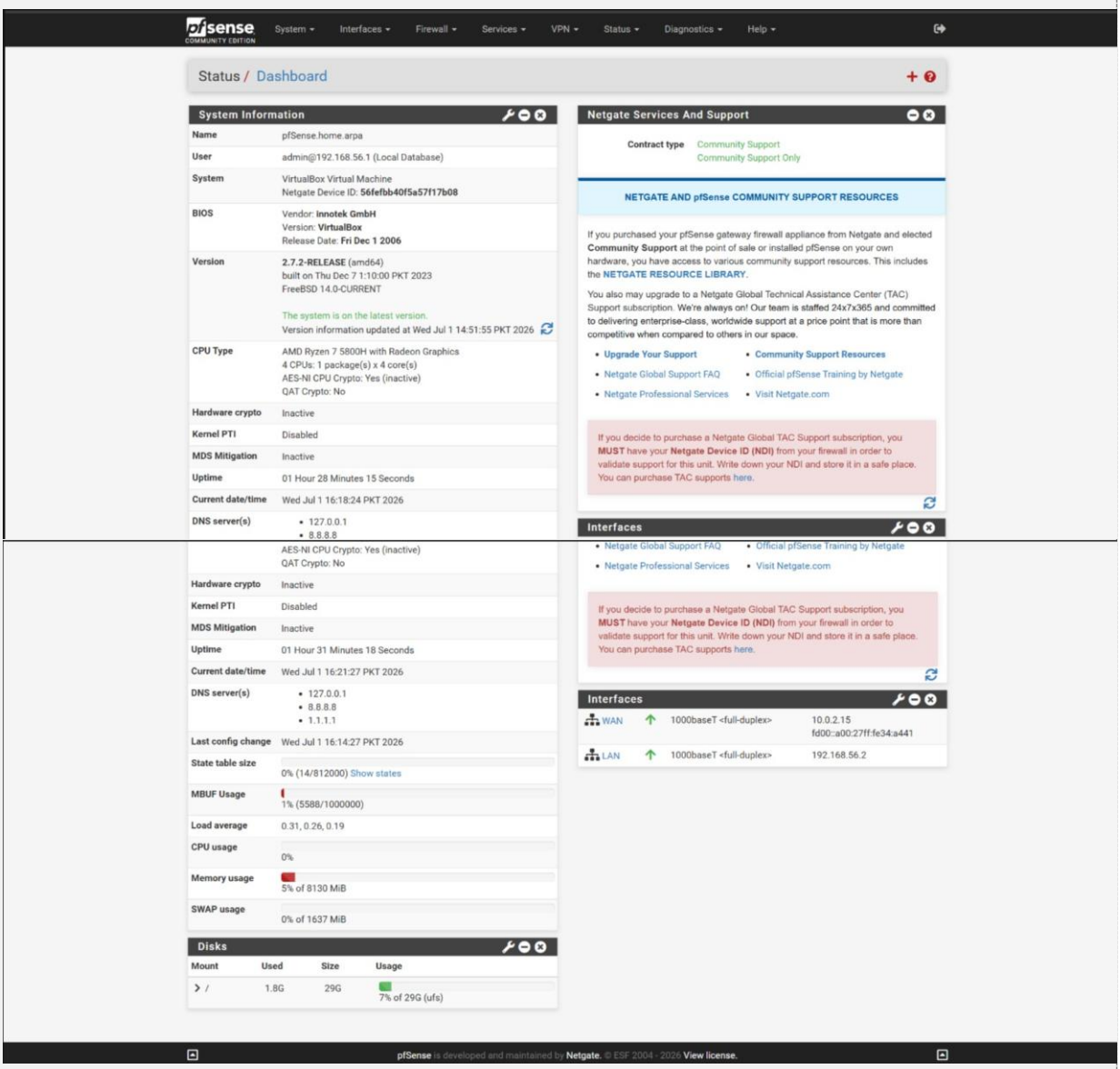
2. pfSense Installation

pfSense Community Edition was installed inside a VirtualBox VM. During setup, a few common issues came up and were resolved, including enabling hardware virtualization (VT-x) on the host, disabling Windows features that conflict with VirtualBox (such as Hyper-V and Memory Integrity), and switching from the newer Netgate installer image to the more stable classic CE installer image, since the Netgate installer failed to connect to its internal setup service.

Once installed, the setup wizard was used to configure the hostname, domain, DNS servers, time zone, and the WAN and LAN interfaces. WAN was set to obtain an address automatically through DHCP (from VirtualBox NAT), and LAN was given a static address on the Host-only network so that other VMs could reach it consistently.



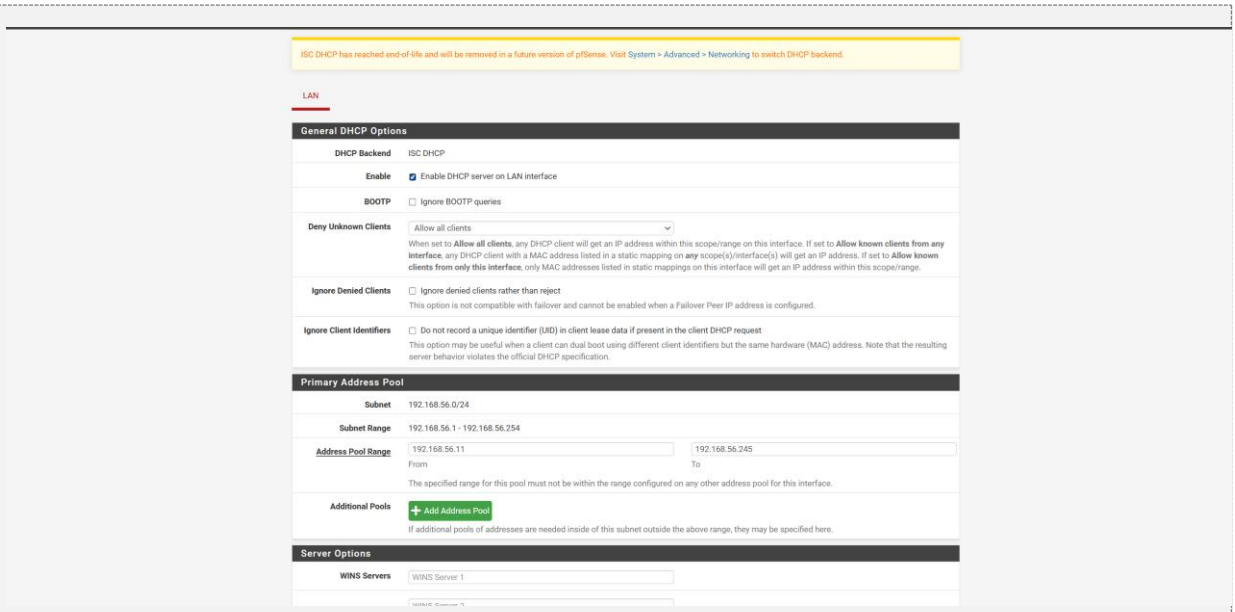
: The pfSense console menu showing WAN and LAN interface IPs



The main pfSense Dashboard page after logging in, showing system information, WAN status, and LAN status.

3. DHCP and Network Settings

The DHCP server on the LAN interface was enabled so that any device connecting to the internal network (such as the Ubuntu client) automatically receives an IP address, a default gateway, and a DNS server pointing to pfSense. This was an important fix during setup, since without a properly configured gateway, LAN devices could reach pfSense itself but could not reach the internet through it.

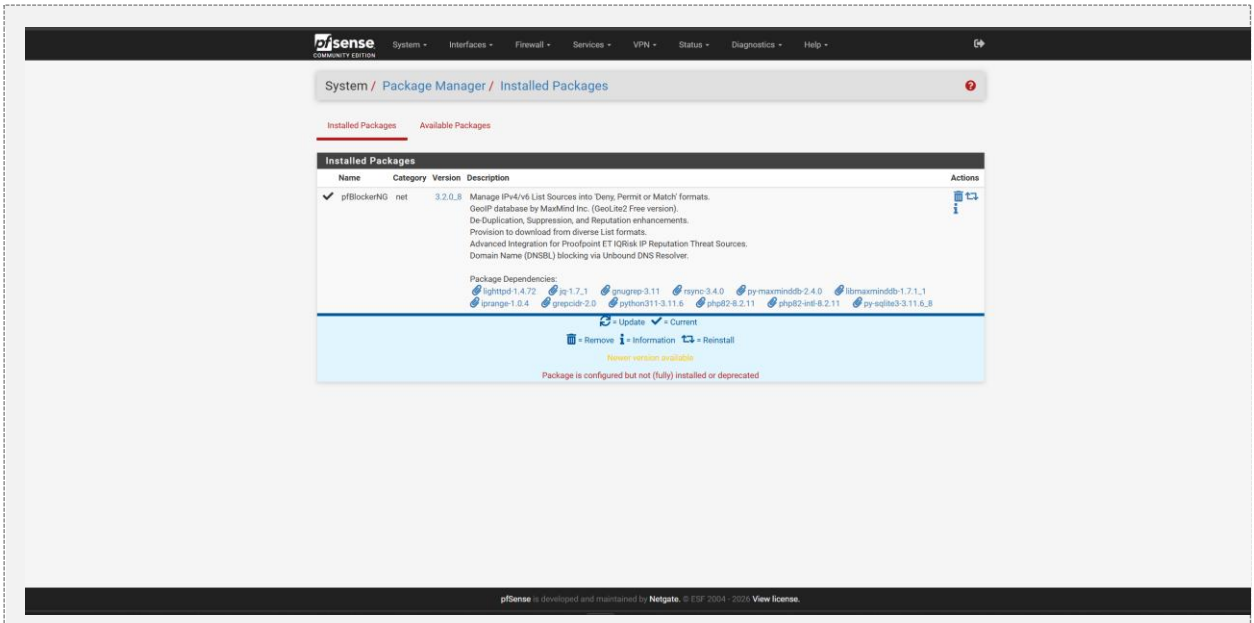


Page showing the enabled checkbox, address pool range, gateway, and DNS server fields filled in.

B. Security Rules Created

1. pfBlockerNG Installation

To implement country-based blocking and domain-based blocking, the pfBlockerNG-devel package was installed through the pfSense Package Manager. pfBlockerNG uses MaxMind's GeoIP database to identify which country an IP address belongs to, and it can also block DNS lookups for specific domains through its DNSBL feature.

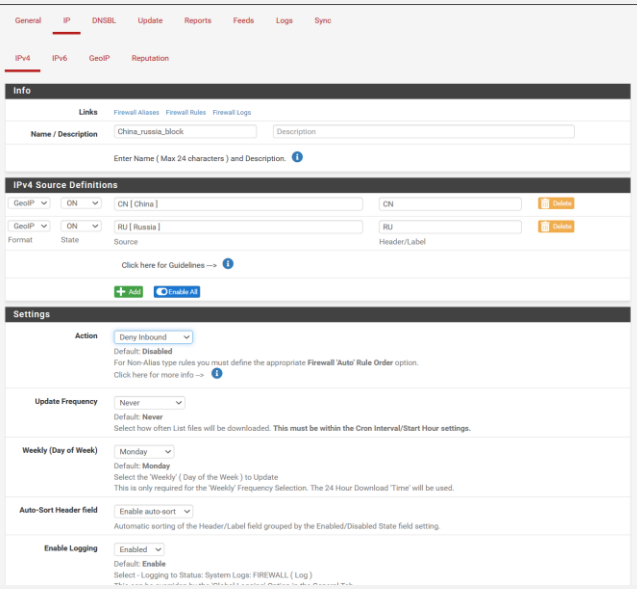


pfBlockerNG installed successfully.

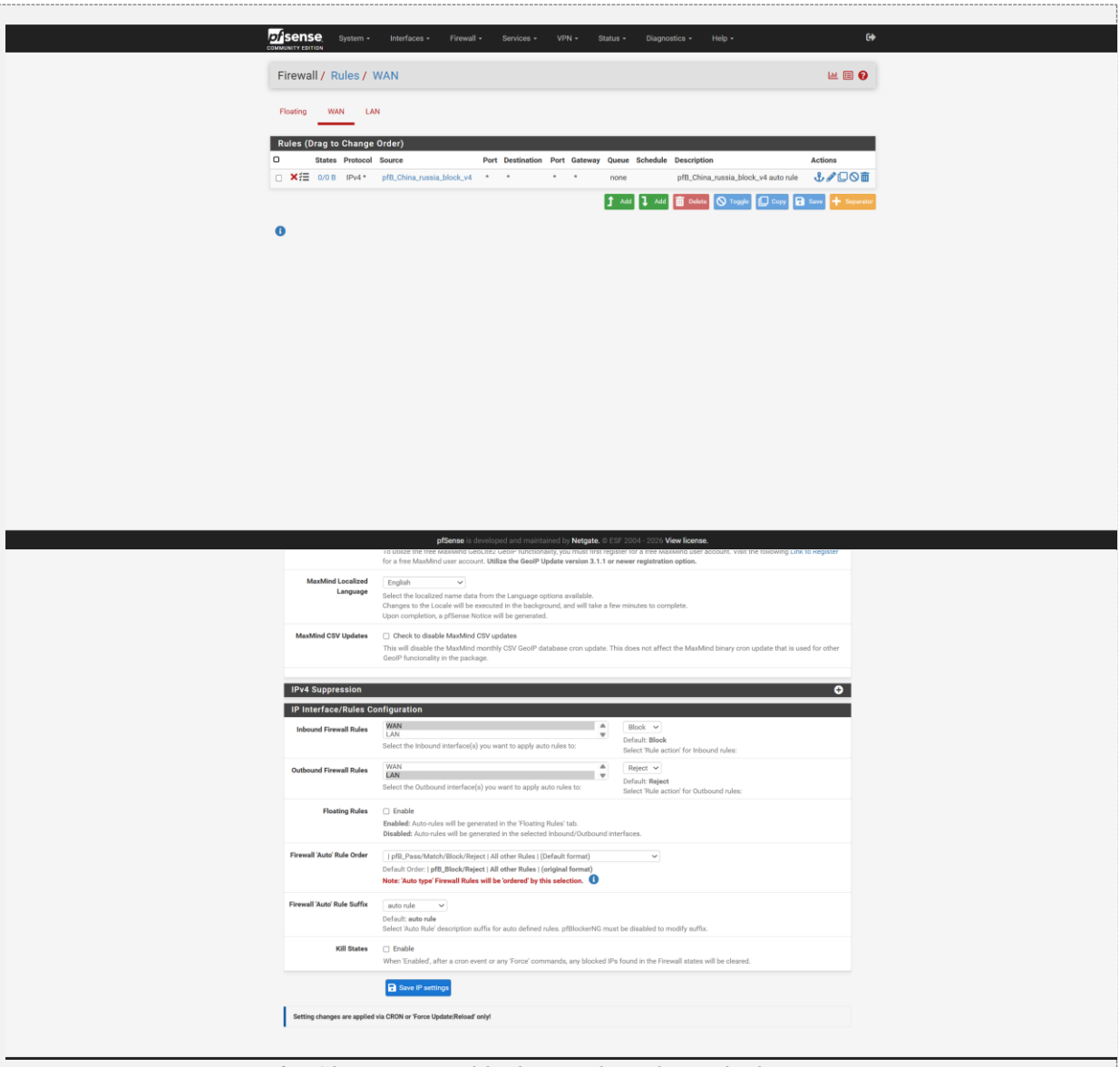
2. Country-Based Blocking (GeoIP)

A custom GeoIP list named 'China_russia_block' was created under Firewall > pfBlockerNG > IP. This list uses the GeoIP source type and is scoped specifically to two country codes: CN (China) and RU (Russia), rather than blocking entire continents. The action for this list is set to Deny Inbound, and it is applied to the WAN interface for inbound traffic and the LAN interface for outbound traffic. This means traffic coming from China or Russia into the network is blocked, and traffic from inside the network trying to reach an IP address in those countries is also blocked.

After the MaxMind GeoIP database was downloaded (using a free MaxMind license key), pfBlockerNG generated a firewall alias containing over 21,000 IP ranges belonging to China and Russia, and automatically created a corresponding block rule on the WAN interface.



IPv4 page showing the China_russia_block entry with CN and RU selected as source definitions.



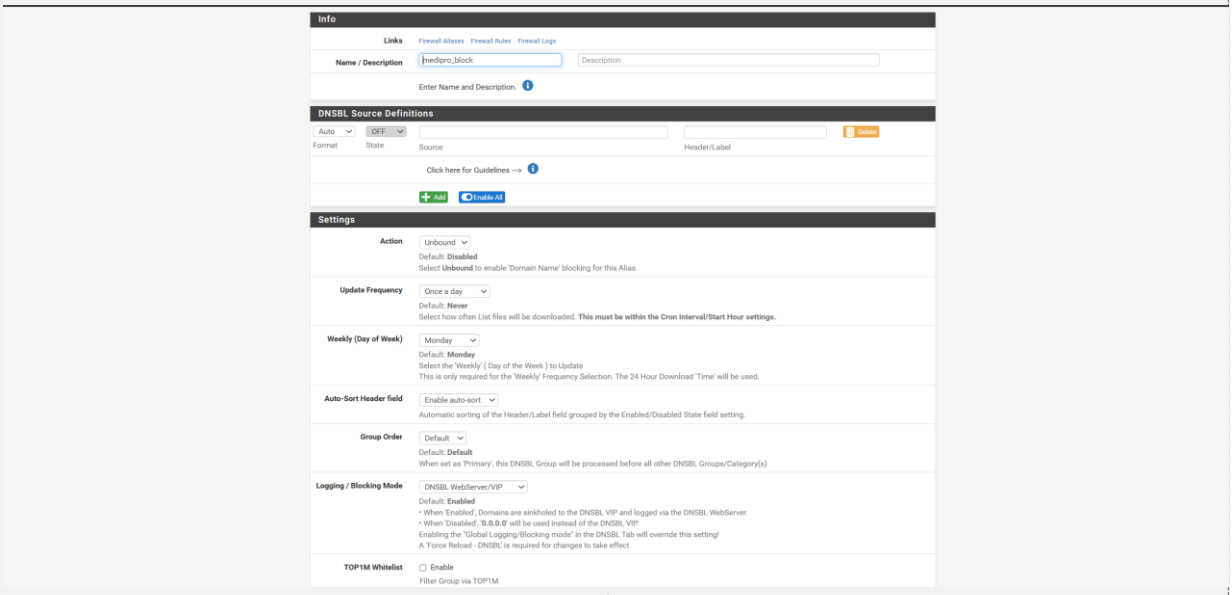
pfB_China_russia_block_v4 rule with its Block action.

3. Website Restriction (DNSBL)

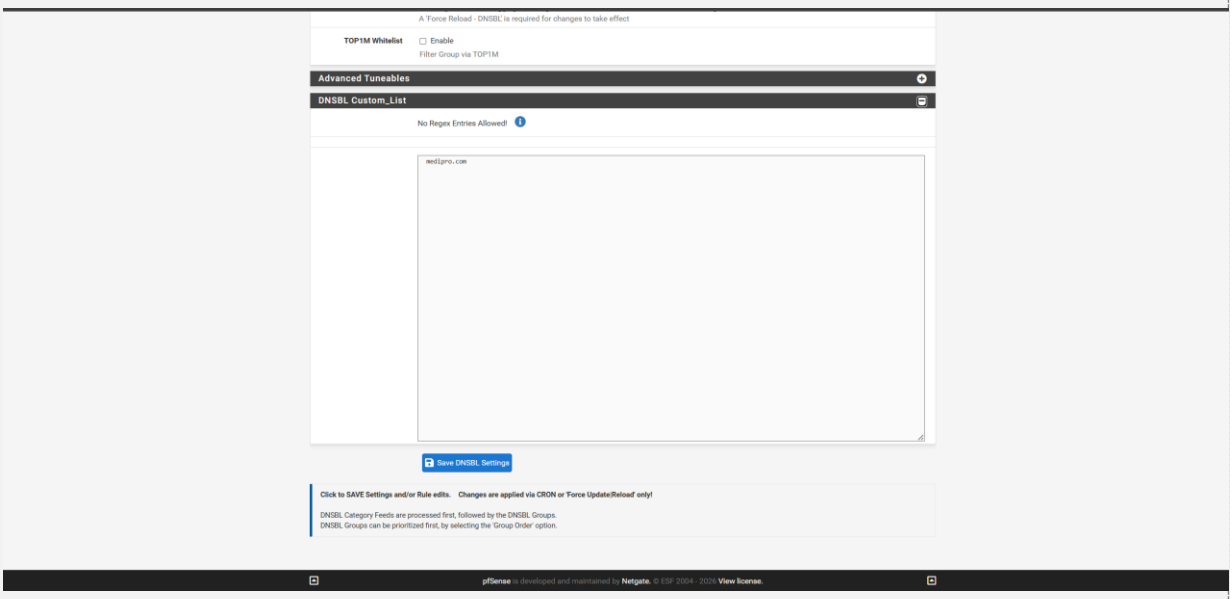
Website blocking was implemented using pfBlockerNG's DNSBL feature, which works at the DNS level rather than the packet level. When a device on the network tries to look up a blocked domain name, pfSense's DNS Resolver intercepts the request and returns the address of an internal sinkhole page (10.10.10.1) instead of the real website address. A custom domain block list was created and tested using a specific domain, which was confirmed to be blocked while other domains such as Wikipedia continued to resolve normally.

One issue encountered during this step was that some domains, including Google and YouTube, are included in pfBlockerNG's default whitelist, which is meant to prevent accidental blocking of

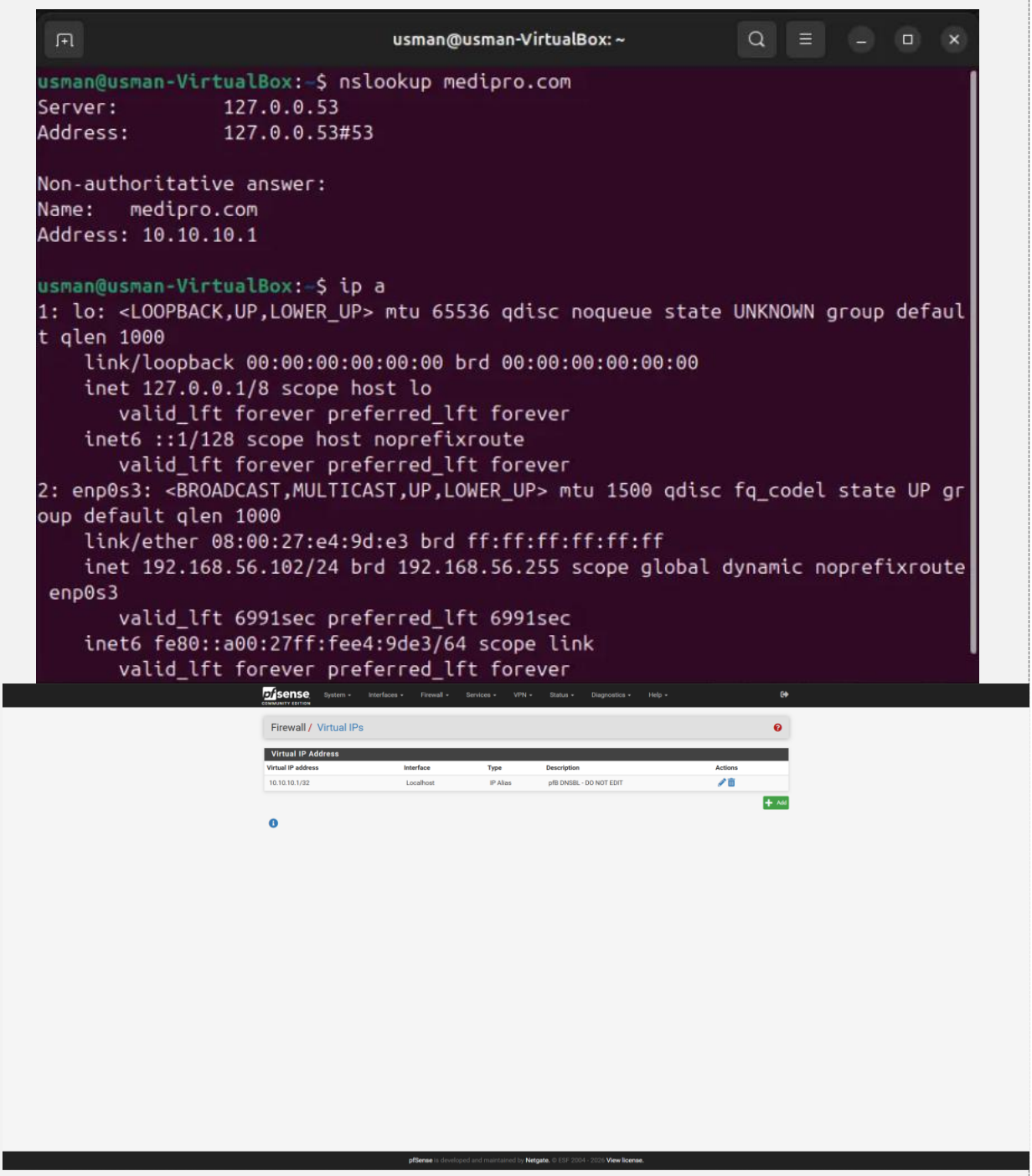
common legitimate services. These entries were removed from the whitelist to allow the intended domain to be blocked successfully.



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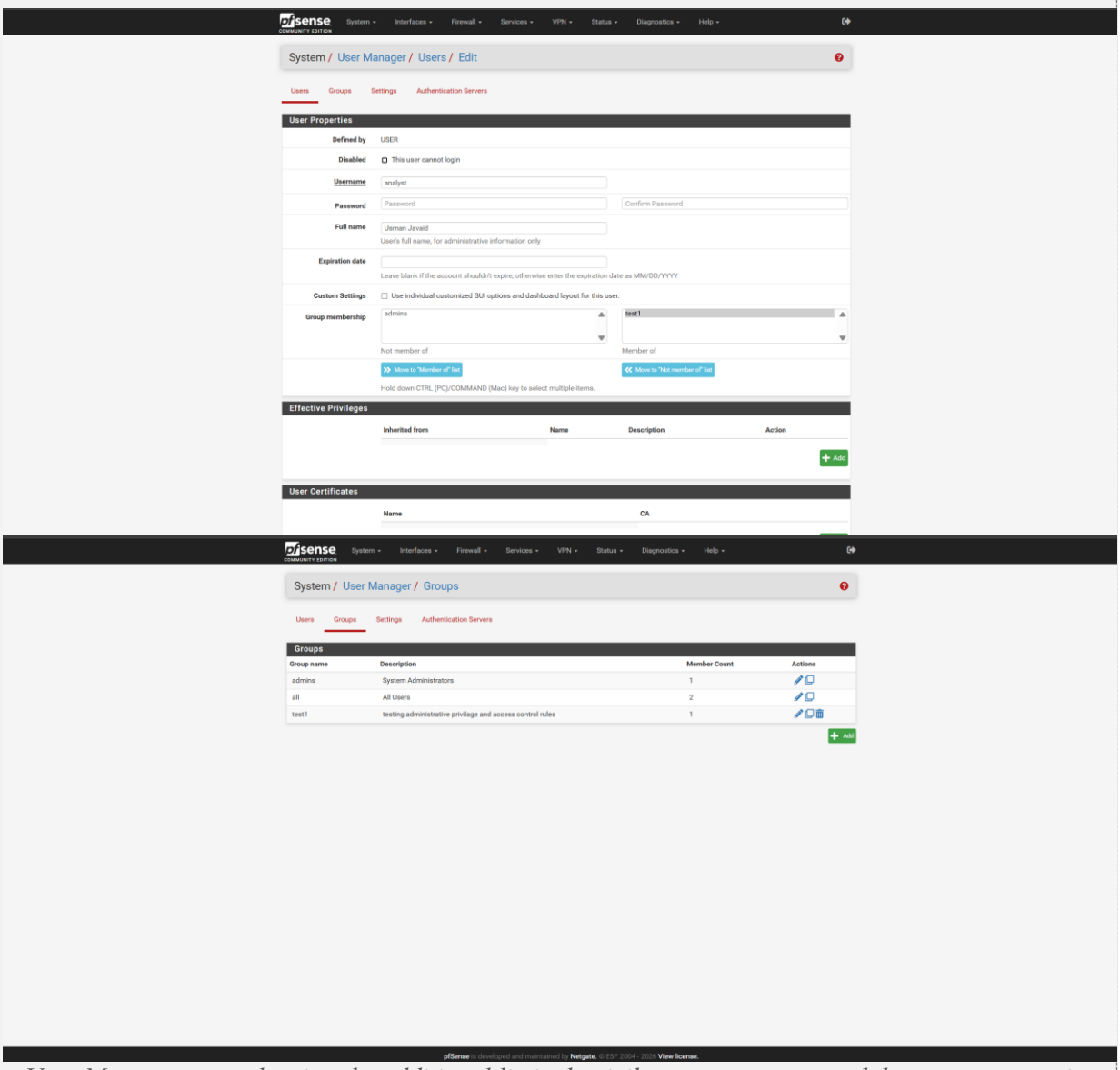
DNSBL page showing the custom domain block entry and its settings



Terminal screenshot from the Ubuntu VM running nslookup on the blocked domain, showing it resolve to 10.10.10.1 instead of the real address.

4. Administrator Access Control

To control administrative access to the firewall, a second, limited-privilege user account was created under System > User Manager, separate from the default full-access admin account. This account was placed in a restricted group with limited privileges, rather than being given full administrative rights. This follows the principle of least privilege, so that not every person who needs to view the firewall also has the ability to change its configuration.



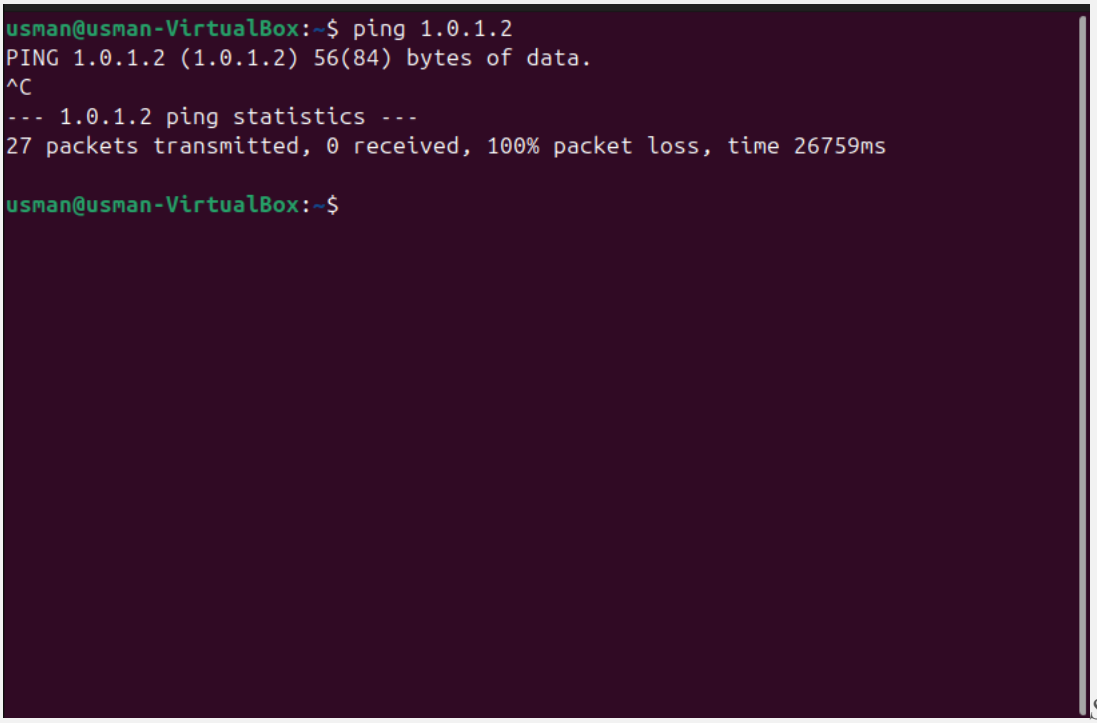
User Manager page showing the additional limited-privilege user account and the custom group it belongs to.

C. Traffic Analysis

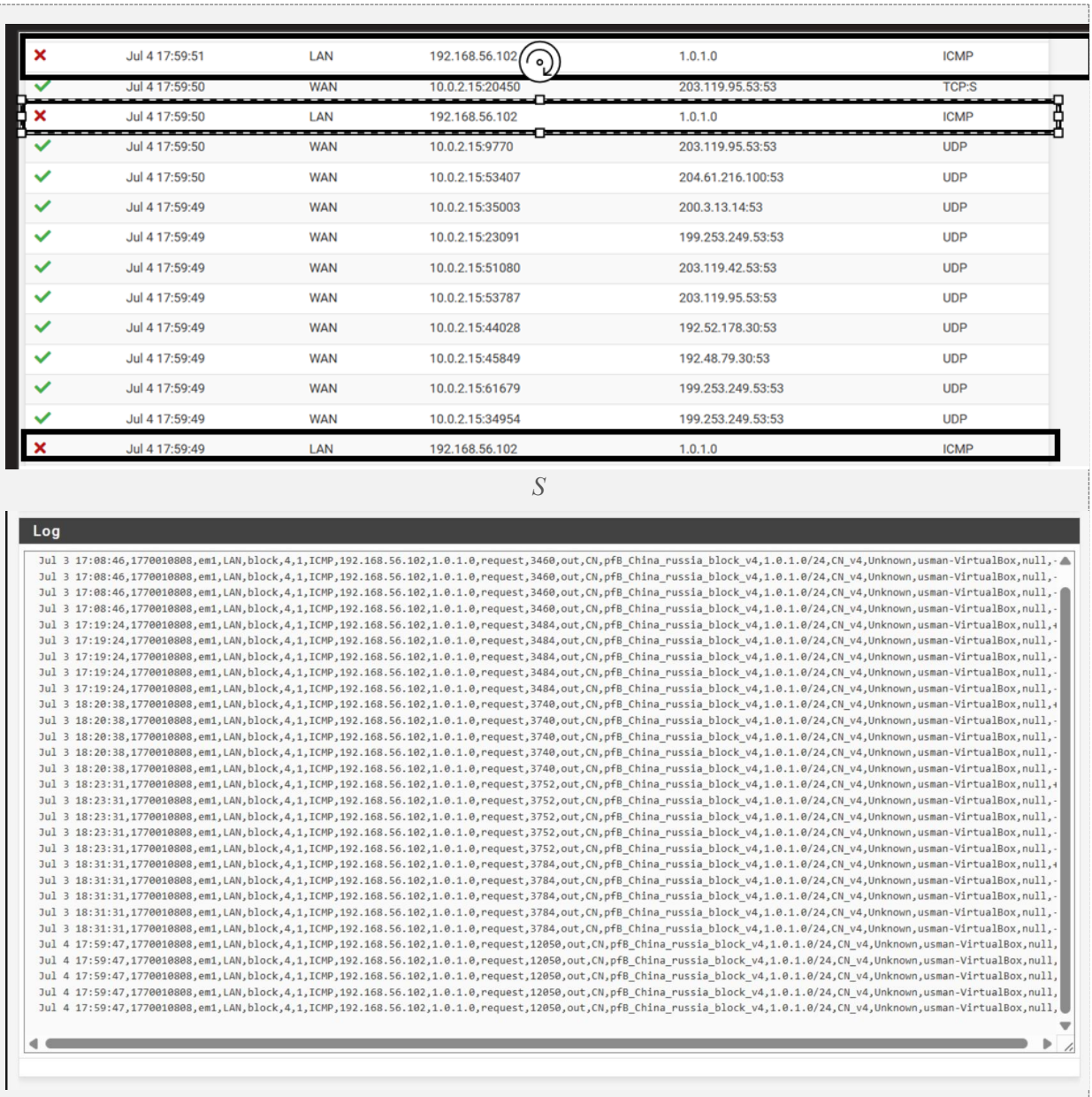
1. Testing the GeoIP Block

To confirm the country block was working, an IP address known to belong to China was pulled directly from pfBlockerNG's own alias file to guarantee it was actually within the blocked range. A ping was then sent to this address from the Ubuntu client. The ping received no replies at all, which is expected behaviour for a silent deny rule, since the firewall drops the packets instead of sending a rejection message back.

This was confirmed further by checking the pfSense firewall log, which showed the blocked packet with full detail, including the source IP (the Ubuntu client), the destination IP, the country code CN, the interface it was blocked on, and the exact rule that matched it (pfB_China_russia_block_v4).



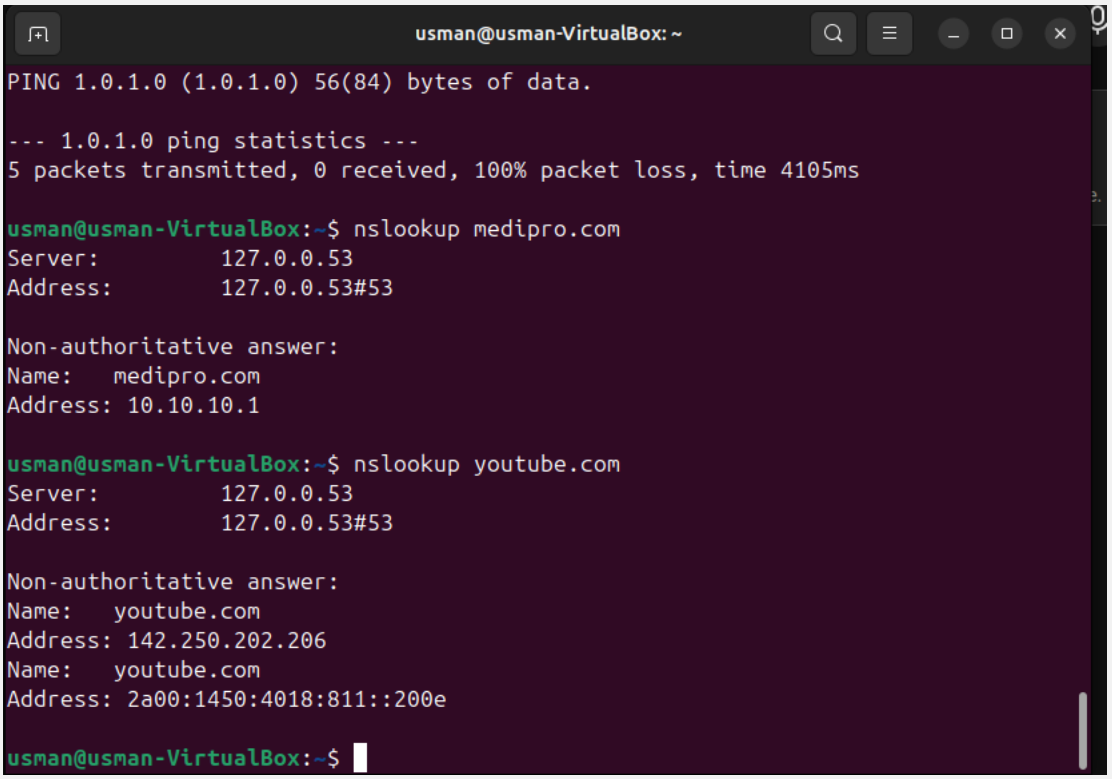
Suggested Terminal on the Ubuntu VM showing the ping command to a China-based IP with 100% packet loss / no replies.



Firewall (Dynamic View) showing the blocked ICMP packet with source, destination, country, and rule name visible.

2. Testing the DNSBL Block

The DNS-based block was tested by running a lookup for the blocked domain from the Ubuntu client. Instead of resolving to the domain's real IP address, the lookup returned the internal DNSBL sinkhole address (10.10.10.1), confirming the block was active. A lookup for an unrelated, non-blocked domain was also run for comparison, and it resolved normally, showing that the block only affects the specific domain that was configured and does not interfere with normal browsing.



nslookup on the blocked domain (returns 10.10.10.1) next to nslookup on a normal domain like wikipedia.org (returns its real IP).

3. Traffic Flow Summary

The table below summarises the traffic tests carried out and their outcomes.

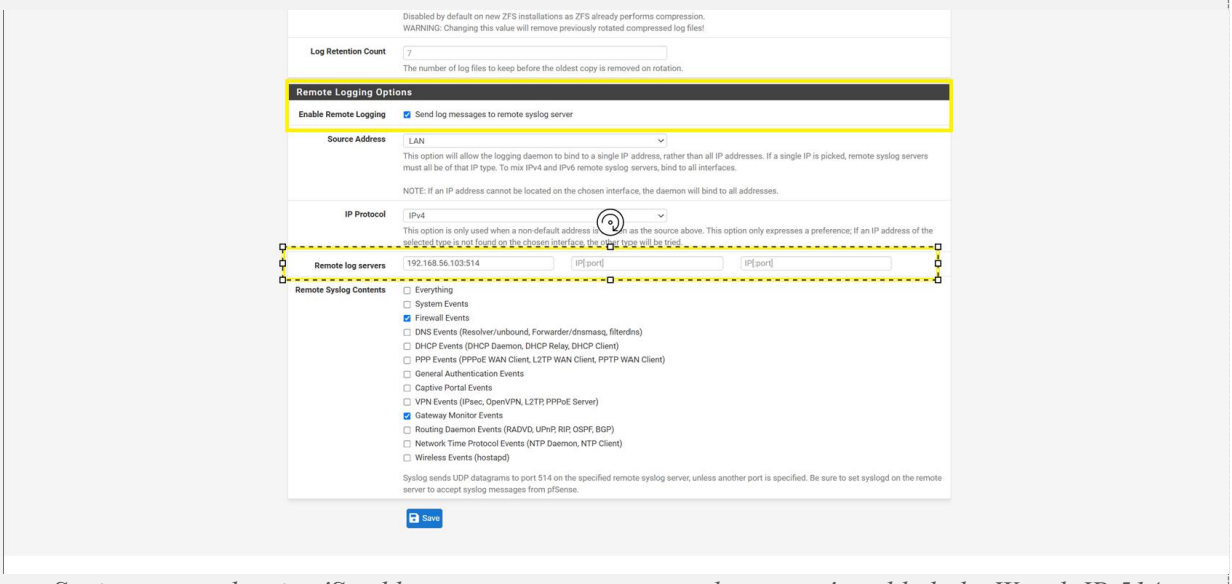
Test	Source	Destination	Result
GeoIP block (China)	Ubuntu client (LAN)	China-based IP (from CN_v4 alias)	Blocked, no reply, logged by pfSense
DNSBL block	Ubuntu client (LAN)	Blocked domain	Resolved to sinkhole IP 10.10.10.1
Control test	Ubuntu client (LAN)	youtube.com	Resolved normally, not blocked

D. Wazuh Alerts and Logs

1. Syslog Forwarding Setup

pfSense was configured to forward its firewall logs to the Wazuh Manager using syslog over UDP port 514, which is the standard port for the syslog protocol. This was set up under Status > System Logs > Settings, with the Wazuh Manager's IP address and port entered as the remote log server, and Firewall Events selected as the log content to forward.

On the Wazuh Manager side, a remote syslog listener was enabled in the manager's configuration file (ossec.conf) to accept incoming syslog traffic from pfSense's LAN IP address on port 514.



Settings page showing 'Send log messages to remote syslog server' enabled, the Wazuh IP:514 entered, and Firewall Events checked.

2. Custom Decoder and Rules

pfSense writes its firewall log entries using a logging process called filterlog, which produces a comma-separated line for every packet that a rule matches. Since this format is specific to pfSense and not recognised by Wazuh out of the box, a custom decoder was written to break this line down into readable fields, including the action taken (block or pass), the interface, the direction of traffic, and the IP version.

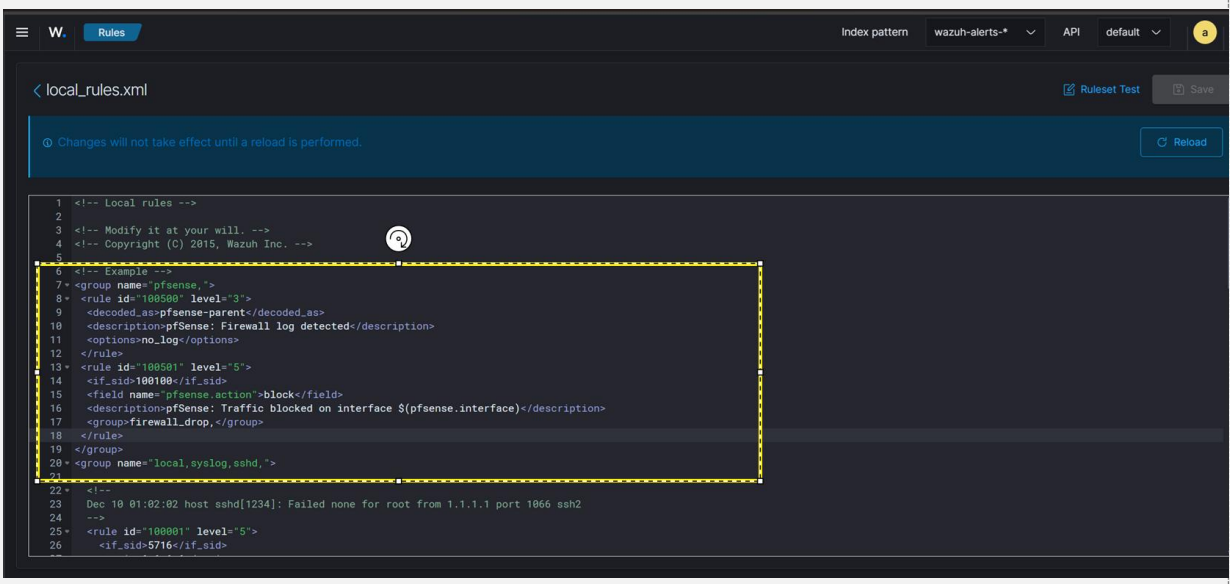
Custom rules were then written on top of this decoder. One rule raises an alert whenever the action field equals 'block', and labels it clearly as a firewall block event. Another rule specifically flags blocked ICMP (ping) traffic. These rules were tested using Wazuh's built-in wazuh-logtest tool before being applied, to confirm they matched correctly.

```
  - Allowed static fields:
  - location   - where the log came from (only on FTS)
  - srcuser    - extracts the source username
  - dstuser    - extracts the destination (target) username
  - user       - an alias to dstuser (only one of the two can be used)
  - srcip      - source ip
  - dstip      - dst ip
  - srcport    - source port
  - dstport    - destination port
  - protocol   - protocol
  - id         - event id
  - url        - url of the event
  - action     - event action (deny, drop, accept, etc)
  - status     - event status (success, failure, etc)
  - extra_data - Any extra data
-->

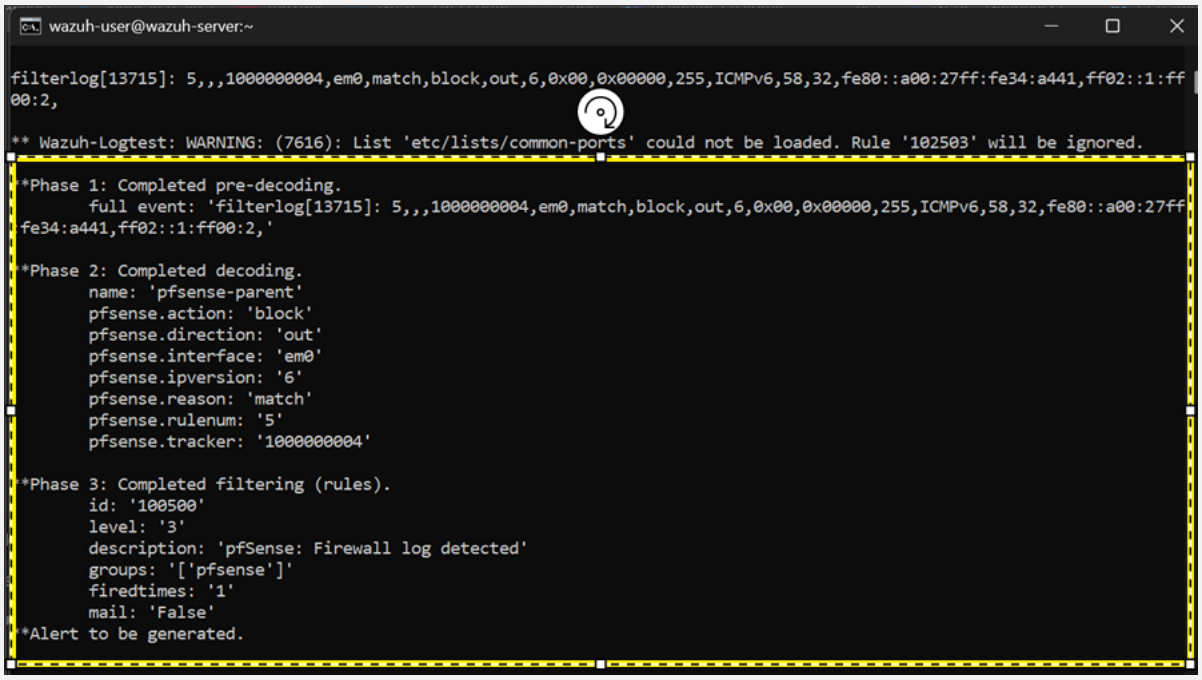
<decoder name="local_decoder_example">
    <program_name>local_decoder_example</program_name>
</decoder>
<decoder name="pfsense-parent">
  <prematch type="pcre2">filterlog\[\d+\]: </prematch>
</decoder>

<decoder name="pfsense-filterlog-ipv4">
  <parent>pfsense-parent</parent>
  <regex offset="after_parent">^(\d+),,,(\d+),(\w+),(\w+),(\w+),(\w+),(\d+)</regex>
  <order>pfsense.rulenum, pfsense.tracker, pfsense.interface, pfsense.reason, pfsense.action, pfsense.direction, pfsense.ipversion</order>
</decoder>
```

The contents of local_decoder.xml on the Wazuh Manager, with the pfsense-parent decoder visible.



The contents of local_rules.xml with the pfsense rule group (block detection).

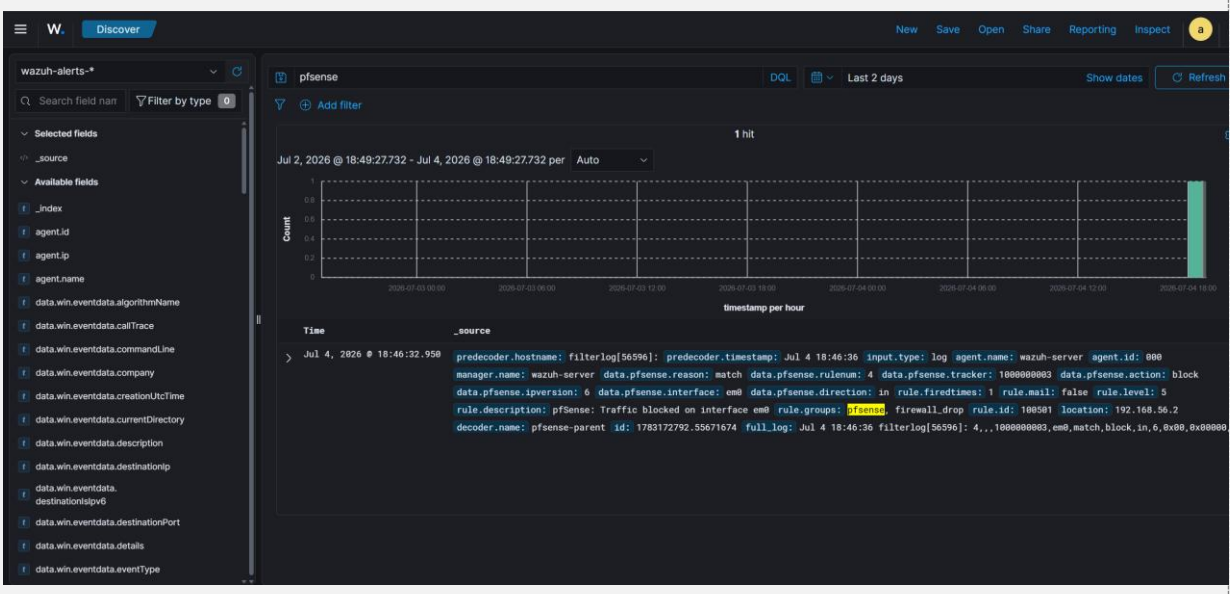


Terminal output of wazuh-logtest showing a pfSense log line being decoded and matched against the custom rule successfully.

3. Alerts in the Wazuh Dashboard

Once the decoder and rules were working, blocked traffic generated during testing appeared as alerts in the Wazuh dashboard under the wazuh-alerts index. Each alert includes the rule description, the severity level, and the relevant fields extracted from the original pfSense log line, allowing an analyst to see at a glance that traffic was blocked and on which interface.

It is worth noting that pfSense's native firewall logging (filterlog) and pfBlockerNG's own internal logging are two separate systems. The pipeline built for this task forwards the native pfSense filterlog output, which includes blocks created by pfBlockerNG rules, since those rules operate as normal firewall rules once generated. pfBlockerNG's separate internal log files (for example, its own DNSBL log) are not forwarded through this same pipeline and would need additional configuration to appear in Wazuh.



Wazuh dashboard (Discover, filtered to wazuh-alerts-) showing an alert with the pfSense block rule description, rule level, and interface/action fields expanded.*

E. Findings and Recommendations

1. Findings

- The GeoIP blocking rule successfully blocked traffic to and from IP ranges belonging to China and Russia, confirmed both by a failed ping test and by a detailed firewall log entry.
- The DNSBL feature successfully blocked a specific domain at the DNS level without affecting unrelated domains, showing that the block was precise and intentional rather than overly broad.
- pfSense's default whitelist inside pfBlockerNG protects a number of common, high-traffic domains (such as Google and YouTube) from being accidentally blocked. This is a sensible default for production use but had to be adjusted for testing purposes in this lab.
- Syslog forwarding from pfSense to Wazuh worked reliably over UDP once the correct IP address, port, and log categories were configured on both ends.
- pfSense's log format is not natively understood by Wazuh and required a custom decoder and custom rules to be written before block events could appear as proper alerts instead of unparsed raw text.
- Two separate logging systems exist on pfSense: the core pf/filterlog system and pfBlockerNG's own internal logs. Only the former was integrated with Wazuh in this task.

2. Recommendations

- Enable logging on pfBlockerNG's own internal lists (IP and DNSBL) in addition to the standard firewall log, so that GeoIP and DNSBL events can also be reviewed directly within pfBlockerNG's own reporting pages, not only through Wazuh.
- Extend the current Wazuh decoder to also extract the source IP, destination IP, and the specific rule alias name (for example pfB_China_russia_block_v4) from each log line. This would let an analyst see exactly which pfBlockerNG list caused a block, directly from the Wazuh alert, without needing to check pfSense separately.
- Forward DNS Resolver (Unbound) logs to Wazuh as a separate log category, so that DNSBL-based website blocks are also visible in Wazuh, not just packet-level GeoIP blocks.
- Review and periodically update the list of blocked countries and domains based on actual traffic patterns and organisational needs, rather than leaving the configuration static.
- For a production environment, switch from UDP to TCP syslog forwarding where possible, since UDP does not guarantee delivery and log entries could be lost under heavy network load.
- Document and version-control the custom decoder and rule files, since they are specific to this pfSense setup and would need to be reapplied if the Wazuh Manager is rebuilt.

This concludes the report for Task 1: Firewall Integration and Monitoring. The firewall was deployed, integrated with Wazuh, and tested against both country-based and domain-based blocking scenarios, with supporting evidence captured through firewall logs, DNS lookups, and Wazuh alerts.